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Species-tree topology impacts the inference of ancient whole-genome duplications across the angiosperm phylogeny

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Abstract

Premise: The history of angiosperms is marked by repeated rounds of ancient whole-genome duplications (WGDs). Here we used state-of-the-art methods to provide an up-to-date view of the distribution of WGDs in the history of angiosperms that considers both uncertainty introduced by different WGD inference methods and different underlying species-tree hypotheses.

Methods: We used the distribution synonymous divergences (K_s) of paralogs and orthologs from transcriptomic and genomic data to infer and place WGDs across two hypothesized angiosperm phylogenies. We further tested these WGD hypotheses with syntenic inferences and Bayesian models of duplicate gene gain and loss.

Results: The predicted number of WGDs in the history of angiosperms (~170) based on the current taxon sampling is largely similar across different inference methods, but varies in the precise placement of WGDs on the phylogeny. K_s-based methods often yield alternative hypothesized WGD placements due to variation in substitution rates among lineages. Phylogenetic models of duplicate gene gain and loss are more robust to topological variation. However, errors in species-tree inference can still produce spurious WGD hypotheses, regardless of method used.

Conclusions: Here we showed that different WGD inference methods largely agree on an average of 3.5 WGD in the history of individual angiosperm species. However, the precise placement of WGDs on the phylogeny is subject to the WGD inference method and tree topology. As researchers continue to test hypotheses regarding the impacts ancient WGDs have on angiosperm evolution, it is important to consider the uncertainty of the phylogeny as well as WGD inference methods.

KEYWORDS

angiosperms, Bayesian, polyploidy, synonymous divergence, synteny, whole-genome duplication

Over 100 years after the discovery of whole-genome duplication (WGD) (Winkler, 1916; Winge, 1917), the number and distribution of ancient WGDs or paleopolyploidization events across the angiosperm phylogeny remains uncertain. Regardless, the high frequency of ancient WGDs in angiosperms has spurred additional questions, such as the association of WGD with the emergence of traits of interest (Jiao et al., 2011; Soltis and Soltis, 2016; Moriyama and Koshiba-Takeuchi, 2018), with the timing of major environmental perturbations (Vanneste et al., 2014; Van de Peer et al., 2017; Wu et al., 2020), and with diversification rate shifts (Vanneste et al., 2014; Tank et al., 2015; Landis et al., 2018). The increasing availability of genomic data has encouraged researchers to develop and improve methods for inferring and placing WGD on phylogenies (e.g., Barker et al., 2008; Thomas et al., 2017; One Thousand Plant Transcriptomes Initiative [OTPT], 2019; Zwaenepoel and Van de Peer, 2019a). Despite innovations in WGD inference methods, few studies have attempted to apply current tools across a broad phylogenetic context (but see OTPT, 2019). Similarly, relatively few studies have explored the influence that species-tree inference and WGD inference may have on each other (Shi, 2013; Struck, 2013; Smith and Hahn, 2022; Xiong et al., 2022) and how species-tree inference shapes our estimation of the history of WGDs in flowering plants. When formulating

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hypotheses for the role of WGD in plant evolution, it is important to consider that the species tree itself is a hypothesis, to which any hypothesized WGDs are nested within. This chain of inference can result in varying degrees of uncertainty that should be carefully considered when testing models of evolution, but this source of uncertainty is typically not addressed in current studies (OTPT, 2019; Mabry et al., 2020; Chen et al., 2023). For these reasons, there is a need for an updated description of the distribution of WGDs across flowering plants that considers the sensitivity of the inferences to underlying uncertainty in tree topology.

Methods for detecting and placing WGDs on phylogenies generally utilize three data types. The simplest of these (1) is to estimate the synonymous divergence (K_s) of duplicated genes in genomic or transcriptomic data (Barker et al., 2008; Ambrose and Purugganan, 2012; Vanneste et al., 2014; Tiley et al., 2018; Sutherland et al., 2024). When plotted as a histogram, WGDs show up in K_s distributions as peaks, and mixture models can be used to ascertain a mean or median sequence divergence among the paralogs (Tiley et al., 2018). The median K_s can then be compared to ortholog divergences with other species to determine whether the WGD is in the common ancestry of the species of interest. Inferences using K_s plots can also use rate corrections that account for variable rates of synonymous substitutions among lineages (Barker et al., 2008, 2009; Shi et al., 2010; Sensalari et al., 2022). (2) If genomic data are available, K_s-based methods are often accompanied with genome structural information that can give further insights into the presence of WGD. By making inter- or intragenomic comparisons, regions where genes occur in the same order—collinearity—can be detected (Coghlan et al., 2005; Tang et al., 2008a). In the case of WGD, entire chromosomes are duplicated and often rearranged, forming groups of collinear genes commonly referred to as syntenic blocks (Renwick, 1971; Passarge et al., 1999; Kellis et al., 2004; Byrne and Wolfe, 2005; Jiao et al., 2014). Other processes can create syntenic blocks, and plant species often experienced more than one WGD, sometimes making it difficult to ascertain whether two species share the same WGD history (Tang et al., 2008a, 2011; OTPT, 2019; Siddiqui and Conant, 2023). Despite these limitations, the availability of data and the accessibility of tools that can quickly generate K_s and syntenic inferences have made both approaches popular (Haas et al., 2004; Barker et al., 2010; Tang et al., 2011; Wang et al., 2012; Zwaenepoel and Van de Peer, 2019b; Henry et al., 2022; Sun et al., 2022).

In addition to methods focused on WGD inference in one or two species at a time, a variety of phylogenetic approaches have been developed to infer and place WGDs in the phylogenetic history of multiple species. When WGD occurs, the large influx of duplicate genes is expected to increase duplication rates along specific branches in the species tree. (3) Several methods have been developed to estimate gene duplications along a phylogeny by counting genes (Rabier et al., 2014; Zwaenepoel and Van de Peer, 2020), using species trees to count duplicate subtrees in gene trees (Li et al., 2015, 2018), mapping clade-specific duplications in gene trees onto a species tree (Yang et al., 2015), and using gene-tree to species-tree reconciliations (Chen et al., 2000; Pfeil et al., 2005; McKain et al., 2016; Thomas et al., 2017; Zwaenepoel and Van de Peer, 2019a). The most recent iterations of these methods account for variation in the background rate of small-scale duplications (SSD) across a phylogeny when testing for the presence of WGD (Zwaenepoel and Van de Peer, 2019a, 2020). These approaches were recently incorporated into WHALE (Zwaenepoel and Van de Peer, 2019a), which considers gene-tree to species-tree reconciliation in a Bayesian context, allowing for the uncertainty of gene-tree topologies to be explicitly modeled. One limitation of WHALE is that it does not detect WGD itself, but rather allows users to test prior WGD hypotheses inferred using other methods. Due to this limitation, the hypothesis generation step is critical and should carefully be considered by researchers.

Here we applied state-of-the-art methods on a sample of over 400 angiosperm species using two different species-tree hypotheses. We made initial WGD hypotheses using a K_s-based approach with KsRates (Sensalari et al., 2022). Using this method, we were able to produce multiple alternative hypotheses for the placement of WGDs in the history of flowering plants. We further assessed the support for these placements using two models available in WHALE: the "critical model" that allows for gene duplication and loss rates to vary among branches, and the "relaxed model" that allows rate heterogeneity both among and within branches (Zwaenepoel and Van de Peer, 2019a). In doing so, we explicitly account for variability in the gene duplication and loss rates across our sampled species. To provide additional support for WGD inferences and placements, we compared patterns of synteny among more than 100 reference genomes. We conducted these analyses using two previously published species-tree topologies (Kumar et al., 2017, 2022; Janssens et al., 2020) to better understand the sensitivity of current methods to topological variation and give a broad overview of WGD history across angiosperms that reflects the sensitivity and variation of current methods for inferring WGDs across different species-tree topologies. Given the increasing availability of genomic data, further exploration into how tree topologies and methods influence WGD inferences is critical as we form and test hypotheses concerning how WGDs may have impacted plant evolution.

MATERIALS AND METHODS

Sequence data sets

We collected 462 genomes and transcriptomes for the flowering plants and outgroups to infer and place WGDs

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across the angiosperm phylogeny. We downloaded 108 publicly available flowering plant genomes from Ensemble, Phytozome, NCBI, and several smaller databases (Appendix S1: Table S1). To fill in unsampled sections of the angiosperm phylogeny we included transcriptomes from the One Thousand Plant Transcriptomes Initiative (OTPT, 2019). Missing gene annotations and inflated species sample sizes can make post hoc testing of WGD hypotheses using WHALE computationally infeasible (Zwaenepoel and Van de Peer, 2019a; Chen et al., 2023). To accommodate these limitations and test prior WGD hypotheses made in the OTPT analyses, we filtered the OTPT data set by only considering species that share previously hypothesized WGD: For each of these previously inferred WGDs, we retained the two descendant species with the highest BUSCO score, along with the next closest outgroup that did not have the WGD in its ancestry. After applying this procedure to the OTPT data set, we went through the data set a second time, and for any family not represented in the post-filtering data set we added back one representative species with the highest BUSCO score. Following this final addition of family representatives, we retained OTPT transcriptomes for 354 species in addition to the 108 reference genomes for subsequent analyses (Appendix S1: Tables S1–S3).

Phylogenetic data sets

We used two different methods to obtain time-calibrated phylogenies for WGD inference with WHALE. We obtained one set of time calibrated phylogenies from TimeTree.org (November 2022), a service that provides tree topologies and species divergence times conglomerated from the literature (Kumar et al., 2017, 2022). To explore how tree topology influences the placement of WGDs in the phylogeny, we used a second time-calibrated phylogeny (Janssens et al., 2020). In contrast to the novel time calibration approach used by TimeTree, the tree of Janssens et al. (2020) is based on a more traditional dating approach using a phylogeny inferred from *matK* and *rbcL* sequences across 36,101 species and BEAST.v1.10 to develop a calibrated tree based on 56 fossil calibrations. Several species in our data set were not present in one or both of the selected trees, so we assigned them to the leaf of the next closest species in the phylogeny, typically a closely related species from the same genus or family (Appendix S1: Tables S2, S3). To make computation time and analyses with WHALE feasible, we split each of these phylogenies into 42 analyses of approximately 10–20 species each (Appendix S1: Tables S2, S3). WHALE cannot test WGD hypotheses at the root of trees, requiring breaks in our larger phylogenies to consider prior WGD hypotheses. To ensure WGDs were on internal branches in each analysis, we use the WGD hypotheses from the OTPT (2019) study. Further, for locations in the phylogeny where the duplications were near analysis breaks, we included an overlap of two or more species between neighboring analyses to avoid making assumptions as to the precise local placement of a given WGD.

Hypotheses for WGD placement

WHALE requires prior hypotheses for the placement of WGD on the phylogeny to test whether paralog retention rates on a given branch in the phylogeny are consistent with WGD. To initially infer WGDs across our phylogenomic data sets, we used the KsRates package to generate K_s plots for each species and conduct rate corrections among species (Sensalari et al., 2022). KsRates forms one or more species trios using different outgroups to correct the ortholog divergence of each species pair in an analysis. For the 42 splits in each of our species trees, we ran KsRates on default settings; however, we allowed the maximum number of outgroups used to form species trios equal to the total number of species in the subtree. We considered a WGD to be shared among species if the median K_s of the paralog peak was greater than the median K_s of the divergence among each species. Multiple alternative hypothesized placements for WGD may be present in each analysis due to variation in rate corrections. Following inferences, we removed any hypothesized WGD that occurred before divergence with the outgroup or had a median $K_s \leq 0.1$ (Tiley et al., 2018) (Appendix S1: Tables S2, S3).

WHALE analyses

We used WHALE (Zwaenepoel and Van de Peer, 2019a) to test the inferences and phylogenetic placements of WGDs hypothesized by our KsRates analyses. WHALE requires a posterior distribution of gene-tree topologies for each gene family in the phylogenetic sample (Zwaenepoel and Van de Peer, 2019a). Gene families were first inferred using OrthoFinder.v2.5.5 with default settings (Emms and Kelly, 2019) and filtered to remove extremely large or small gene families using orthofilter.py available from WHALE (Zwaenepoel and Van de Peer, 2019a). Gene families that passed the filtering step were aligned using Prank.v.170427 and analyzed with MrBayes.3.2.7a to generate a posterior distribution of gene trees under an LG + GAMMA model of sequence evolution (Ronquist et al., 2012; Emms and Kelly, 2019; Löytynoja, 2021). Each posterior distribution of trees was generated from 11,000 trees sampled every 10 generations from a model that we ran for 110,000 generations. From these distributions of trees, we produced an amalgamated likelihood estimation (ALE) object using ALEobserve, discarding the first 1000 trees as burn-in (Szöllősi et al., 2013). Finally, we used the ccddata.py and ccdfilter.py scripts available from WHALE to filter out gene families with a high degree of uncertainty in their gene-tree topologies (Zwaenepoel and Van de Peer, 2019a). (Data are available from GitLab repository: https://gitlab.com/barker-lab/ajb_special_issue_2024.git.).

The rate of small-scale gene duplication and loss can vary across the phylogeny and impact the inference of

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whole-genome duplication events (Zwaenepoel and Van de Peer, 2019a; Chen et al., 2023). To accommodate this variation in duplication and loss rates, we ran two WHALE models on each 10–20-species analysis across the TimeTree (Kumar et al., 2017, 2022) and Janssens et al. (2020) topologies. The first of these—the "critical" model—allows the rate of gene duplication and loss to vary among branches, but the duplication and loss rates are constrained to be equal within branches. A second "relaxed" model allows the duplication and loss rates to be unconstrained and vary independently among and within branches, representing more biological realistic processes, with the trade-off of increased parameterization. Following Chen et al. (2023), we utilized a Beta(3,1) prior distribution for the number of genes at the root of each tree (η) for both of these models. In the critical model, the mean duplication-loss branch rate (r) was set to an improper flat prior (sensu Bayesian statistics), and each branch rate (λ,μ) was assumed to follow a multivariate Gaussian prior with a standard deviation (σ) from an exponential prior with mean equal to 0.1. In the relaxed model, we allowed log-duplication-loss rates (r) to vary independently following a bivariate normal prior with mean of 0 and standard deviation of 1. Within branches, we assumed a Uniform(−1,1) prior for the correlation coefficient (ρ) between duplication and loss rates and an exponential prior with mean of 1 for the standard deviation of branch rates (τ). For both models we sampled the posterior 500 times, requiring the ESS to reach a value of 100 or greater (average 513) (supplemental data available from GitLab repository). Finally, for each hypothesized WGD placement, we calculated the log10 Bayes factor of the marginal posterior distribution of the retention rate parameter (q) under the critical and relaxed models (supplemental data available from GitLab repository). A given WHALE model supported a WGD hypothesis if the retention rate (q) along the hypothesized branch was significantly different from zero, as indicated by a Savage-Dickey based log10 Bayes factor less than −2.

Syntenic analyses

In addition to WHALE, we used syntenic inferences to further test WGD hypotheses proposed by KsRates. To detect collinearity between the genomes in our data set, we used the python distribution of MCScan in a pairwise fashion (Tang et al., 2008b). For each WGD hypothesis we compared the syntenic depth ratio between ingroup species and outgroup species up to two nodes away from the WGD (supplemental data available from GitLab repository). For each ingroup species, we considered the presence of WGD supported if 50% or more of the syntenic depth comparisons with the outgroup species had a syntenic depth ratio greater than 1:1. The placement of the WGD on the phylogeny was then considered supported if all species daughter to it in the phylogeny met the criterion for the syntenic depth ratio test described. Finally, we used ete3.v3.1.3, matplitlib.v3.8, and seaborn.v0.13.0 to visualize syntenic bidirectional comparisons as a heat map arranged following Janssens et al. (2020) tree topology (Hunter, 2007; Huerta-Cepas et al., 2016; Waskom, 2021).

RESULTS

K_s-based WGD hypotheses

Using KsRates, we inferred 246 and 244 hypothesized WGDs on the TimeTree (Kumar et al., 2017, 2022) and Janssens et al. (2020) topologies, respectively (Appendix S1: Tables S2, S3). Many of these placements represent the same duplication event shifted up or down one or more nodes, depending on which species was used for the rate correction and how KsRates fit mixture models to the paralog distributions. For example, in analysis A1 of the TimeTree topology, the previously described ARTHβ duplication was variously placed to both include and exclude *Limnanthes douglasii* (Figure 1; Appendix S1: Tables S2, S3). These alternative placements result from changing the outgroup species in different rate corrections. For example, in our analysis of *Batis maritima*, the ARTHβ WGD paralog median K_s (1.98) was placed after the rate-corrected divergence with *L. douglassii* (K_s = 2.05) (supplemental data available from the GitLab repository). Alternatively, in our *Salvadora persica* analysis, the ARTHβ paralog median K_s is 1.94 and occurs before the rate corrected divergence with *L. douglassii*

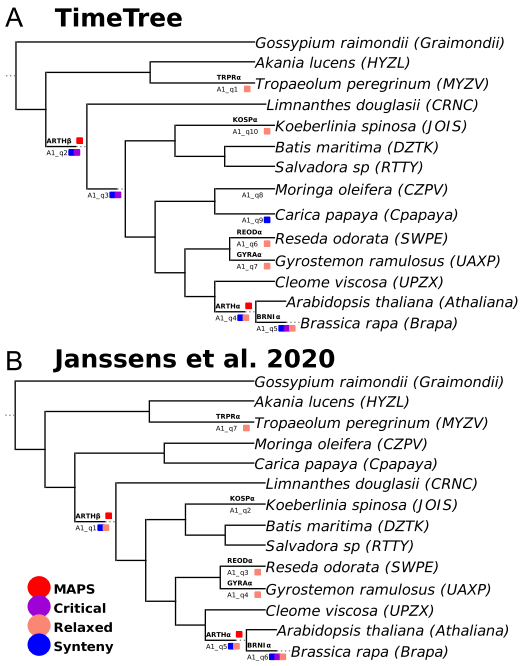


FIGURE 1 Comparison of support from KsRates ("analysis"_"WGD node"), syntenic (blue), WHALE critical (purple), relaxed model (peach), and MAPS (red) for the placements of WGD within the Brassicales under the (A) TimeTree and (B) Janssens et al. (2020) tree topologies. The duplication naming convention from OTPT (2019) was used.

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($K_s = 1.49$) (supplemental data available from the GitLab repository). Similarly, we observed that the mixture models fit by KsRates occasionally placed the distributions of overlapping WGD closer to each other than in prior publications. For example, for *Brassica rapa*, KsRates inferred a median K_s of 0.31 and 0.35 for the BRNIα and ARTHα WGDs (Appendix S1: Tables S2, S3). Previous analyses placed these duplications at median K_s of ~0.3–0.4 and ~0.6–0.7, respectively (Barker et al., 2009; Wang et al., 2013; Kagale et al., 2014; Jeong et al., 2016). All of these model differences were further exacerbated by topological disagreements between the two time-calibrated phylogenies. Continuing the example from analysis A1, *Carica papaya* and *Moringa oleifera* were placed at a shallower node in the TimeTree topology compared to the Janssens et al. (2020) topology, resulting in disagreement on whether they shared the ARTHβ duplication or not (Figure 1). This shallower placement disagrees with the current understanding of the evolutionary history of Brasicales (Barker et al., 2009; Edger et al., 2018; Mabry et al., 2020; Walden et al., 2020), whereas the Janssens et al. (2020) topology is more consistent with previous work in the group, as are the Brassicaceae WGD inferences.

WGD hypothesis testing with WHALE

We used WHALE to assess alternative hypotheses for the placement of WGD in the angiosperm phylogeny using paired analyses in the TimeTree (Kumar et al., 2017, 2022) and Janssens et al. (2020) phylogenies (Zwaenepoel and Van de Peer, 2019a). We conducted 42 analyses of an average species number of 12 for both the TimeTree (Kumar et al., 2017, 2022) and Janssens et al. (2020) species trees. For each analysis, we used an average of 9754 orthofamilies (4804–12,180; median = 9770) as input data for WHALE (supplemental data available from the GitLab repository). We used two models to assess our hypotheses, one that constrained rates of duplication and loss to be equal within branches but vary among branches (the critical model), and another that allowed duplication and loss rates to vary among and within branches (the relaxed model; Zwaenepoel and Van de Peer, 2019a; Chen et al., 2023). Across all of our analyses, the median parameter ESS was 432 (513 average), suggesting a good approximation of the posterior using both models (supplemental data available from the GitLab repository). In the TimeTree topology, the critical WHALE model supported the inference and placement of 56 of the 246 (~23%) hypothesized placements made by KsRates (Figure 2; Appendix S1: Table S4; Appendix S2: Figures S2–S85). The relaxed model supported a much larger fraction of hypothesized WGD inference (167, ~65%) (Figure 2; Appendix S1: Table S4; Appendix S2: Figures S2–S85). We observed a similar pattern in the Janssens et al. (2020) tree topology, where 47 (~19%) and 154 (~63%) of the 244 hypotheses were supported by the critical and relaxed models respectively (Figure 2; Appendix S1: Table S4; Appendix S2: Figures S2–S85). In total, 173 and 159 WGD

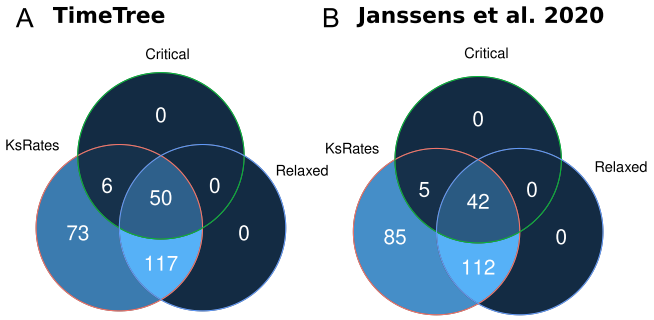


FIGURE 2 Venn diagrams of the total WGD hypotheses proposed by KsRates under the (A) TimeTree and (B) Janssens et al. (2020) topologies and the associated support from WHALE relaxed and critical models.

hypotheses in the TimeTree and Janssens et al. (2020) analyses were supported by at least one of the WHALE models (Figure 2; Appendix S1: Tables S4, S5; Appendix S2: Figures S2–S85).

KsRates and WHALE supported the inference of a similar number of WGDs in both the TimeTree (Kumar et al., 2017, 2022) and Janssens et al. (2020) topologies. However, the placements of these WGDs may represent different evolutionary hypotheses because topological differences can shift which species are inferred to share a WGD event. It can be difficult to disentangle whether a disagreement about the precise placement of WGD by WHALE under both topologies is due to model differences or topological non-equivalency between species trees. However, if a putative WGD is in the shared ancestry of an identical collection of species across different species trees, then we may consider the WGD placements to agree among those analyses. To make such a comparison between the TimeTree and Janssens et al. (2020) phylogenies, we compared the overlap of species descending from each WGD placement supported by WHALE. Between both sets of analyses across the two phylogenies, 124 of the hypothesized WGDs were present in identical species, representing ~72% and ~78% of hypotheses supported by WHALE models in the TimeTree and Janssens et al. (2020) trees respectively (Appendix S1: Tables S4, S5; Appendix S2: Figures S2–S85). Of the remaining hypotheses in the TimeTree topology, 22 were unique and represented less than a 50% overlap with any WGD hypothesis in the Janssens et al. (2020) topology. A large number (13) of these TimeTree-specific hypotheses were WGDs inferred to be present in only one species in the phylogeny (Appendix S1: Tables S4, S5; Appendix S2: Figures S2–S85). A similar pattern was observed in the Janssens et al. (2020) topology, with the presence of 24 topology-specific WGD hypotheses, 18 of which were species-specific (Appendix S1: Tables S4, S5; Appendix S2: Figures S2–S85). Notably, for 79% (33) of these unique species-specific WGDs, KsRates either did not infer WGD at all or placed the WGD deeper in the phylogeny of the alternative tree.

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Synteny-based inferences

We used additional syntenic information from the 108 genomes included in our analyses to further test hypothesized WGDs. Bidirectional intraspecific syntenic analyses were conducted with a pythonic version of MCscan (Tang et al., 2008b). MCscan classifies the syntenic depth ratio between two species, with higher syntenic depth ratios being indicative of the presence of a WGD. We visualized these inferences as a heatmap organized with the Janssens et al. (2020) topology (Figure 3; Appendix S2: Figure S1). In the case of WGD presence, we expect higher syntenic depth ratios (brighter colors) in a band along the *x*-axis for all species sharing a given WGD and lower syntenic depth ratios (darker colors) vertically following up the *y*-axis in the same species. We highlighted an example for the *Brassica specific* mesohexaploidy in Figure 3 and Appendix S2: Figure S1 (Barker et al., 2009; Wang et al., 2013; Kagale et al., 2014; Jeong et al., 2016). To more formally test each of our hypothesized WGD placements, we selected all species with reference genomes up to two nodes up from the focal branch as outgroup species. For each species daughter to the hypothesized WGD position, we considered the WGD to be supported in the ancestry of that species if at least 50% of the outgroup species comparisons had a syntenic depth ratio greater than 1:1. We considered the inference of a WGD on a given branch supported if all daughter species passed the syntenic comparisons to the outgroup species. Due to limited genomic information, we were only able to test 74 and 75 WGD hypotheses under the TimeTree (Kumar et al., 2017, 2022) and Janssens et al. (2020) tree topologies (Appendix S1: Tables S4, S5; Appendix S2: Figures S2–S85). Syntenic depth inferences supported 61 WGD hypotheses from KsRates in the TimeTree topology, 23 of which neither WHALE model supported (Appendix S1: Tables S4, S5; Appendix S2: Figures S2–S85). Syntenic depth analyses supported 55 of the KsRates inferences of WGD in the Janssens et al. (2020) tree topology, 18 of which neither WHALE model supported (Appendix S1: Tables S4, S5; Appendix S2: Figures S2–S85).

Merged tree inferences

We represented WGDs in multiple analyses due to their inherent overlap in our experimental design. We merged these inferences into the larger Janssens et al. (2020) tree to estimate the total number of supported WGD hypotheses across the entire phylogeny (Figure 4). We did not merge inferences for the TimeTree (Kumar et al., 2017, 2022) phylogeny because generating a single TimeTree with all species significantly changes the newly generated tree topology due to the dating method used by TimeTree, yielding relationships inconsistent with the TimeTree phylogenies used in our WHALE analyses (unpublished

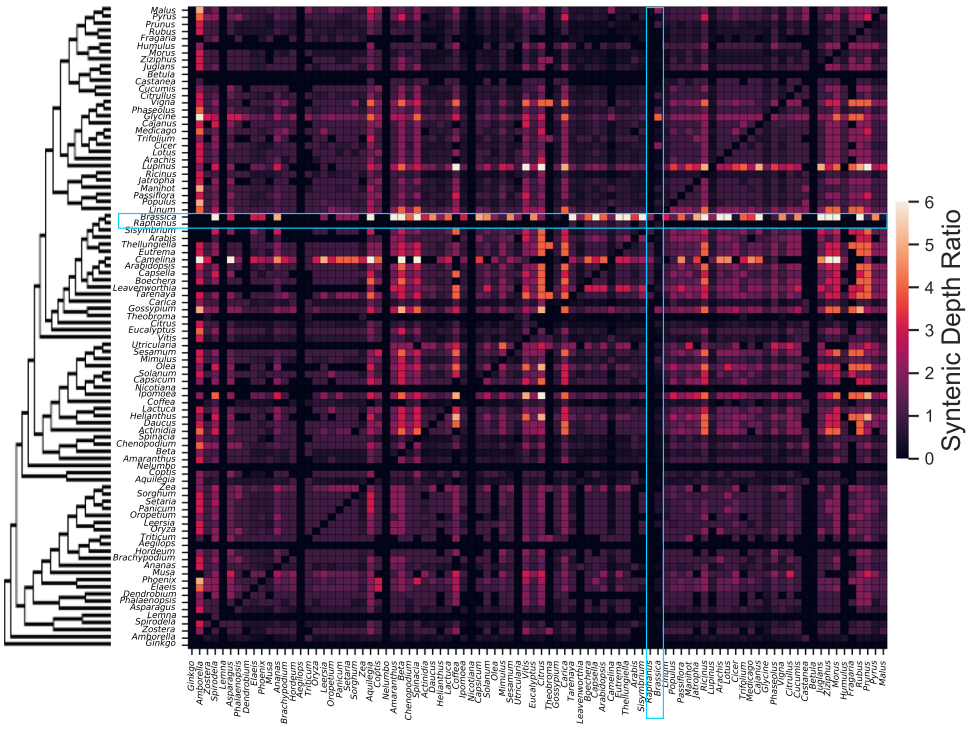


FIGURE 3 Heat map of the bidirectional syntenic depth inferences made by MCScan for each genera. The *y*-axis represent in-group genera analyzed, in the order of the Janssens et al. (2020) tree topology. The *x*-axis represents the outgroup genera in the comparison arranged in the same order as the *y*-axis. We have highlighted an example of the expected pattern of syntenic depth ratios consistent with WGD in blue boxes. Our example shows the *Brassica* mesohexploidy having a consistently high syntenic depth ratio compared to outgroup species and a 1:1 or lower syntenic depth ratio when used as the outgroup for most other species.

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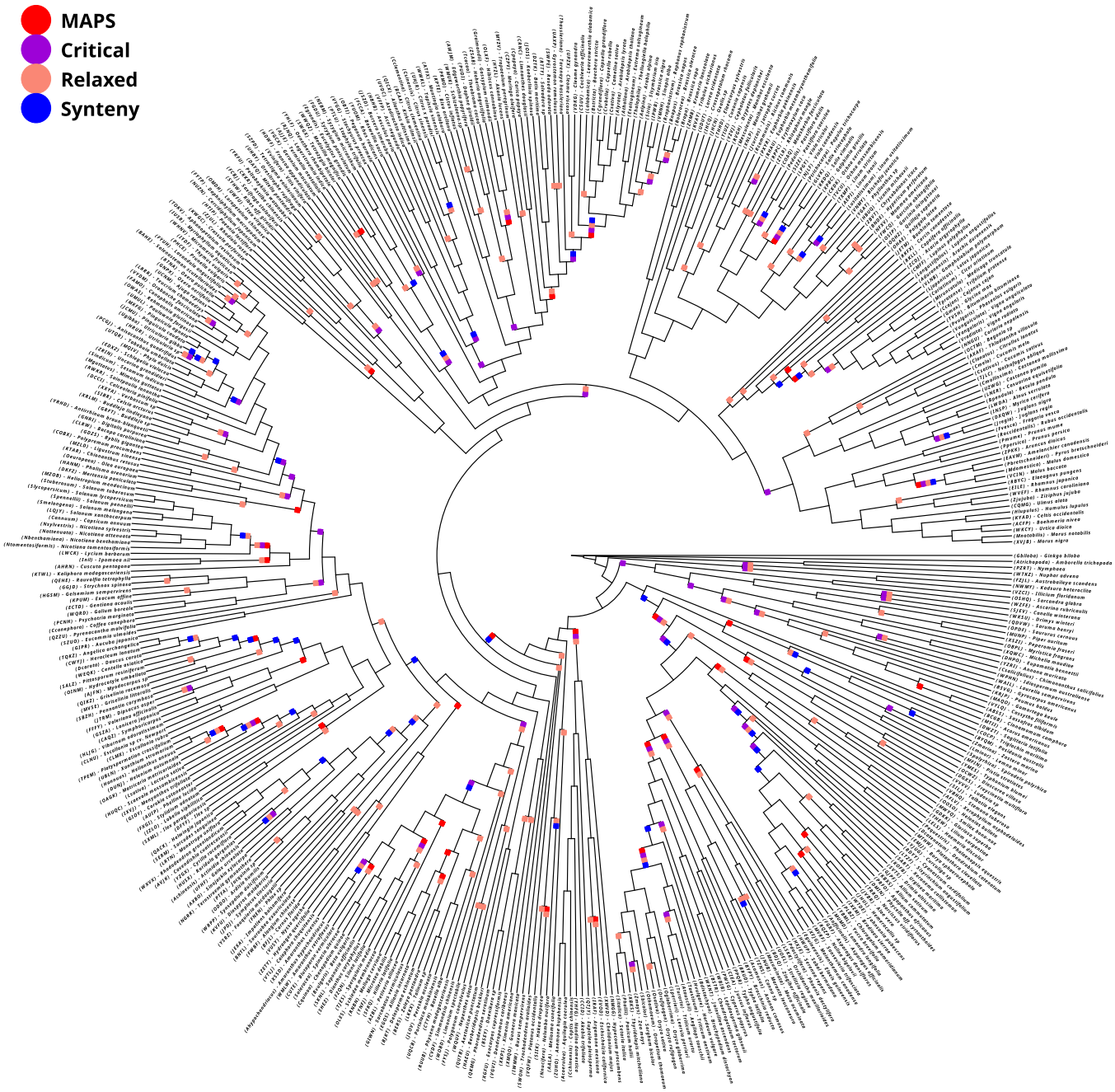


FIGURE 4 Final merged Janssens et al. (2020) phylogeny annotated with hypothesized WGD placements supported by synteny, WHALE models, or MAPS. MAPS inferences were based on the equivalent MRCA for species inferred to share the proposed WGD in the OTPT (2019) analysis. Annotations are as follows: MAPS (red), WHALE critical (purple), relaxed model (peach), and the syntenic depth test (blue).

data). To merge these inferences, we considered two hypothesized WGDs in different analyses as equivalent if there was only one WGD on the branches between the most recent common ancestor (MRCA) nodes of species sharing the duplication in each analysis. Following the merger, a total of 170 WGD hypotheses made by KsRates were also supported by WHALE, 32 of which syntenic information further supported (Figure 4; Appendix S1: Table S6). Our results are similar to prior work from OTPT (2019) that found 179 WGDs in the angiosperms using K_s information, 38 of which were also supported by the multitaxon paleopolyploidy search algorithm (MAPS) from the OTPT (2019) analyses; MAPS is a tool that uses species trees to count duplicate subtrees in gene trees to infer WGD (Li et al., 2015; Li and Barker, 2020). Although the total number of WGDs inferred in OTPT (2019) was similar to our inferences, their placement was not identical. Ten of the placements supported by MAPs did not have a phylogenetically equivalent branch in the Janssens et al. (2020) tree or were not tested due to KsRates not proposing their

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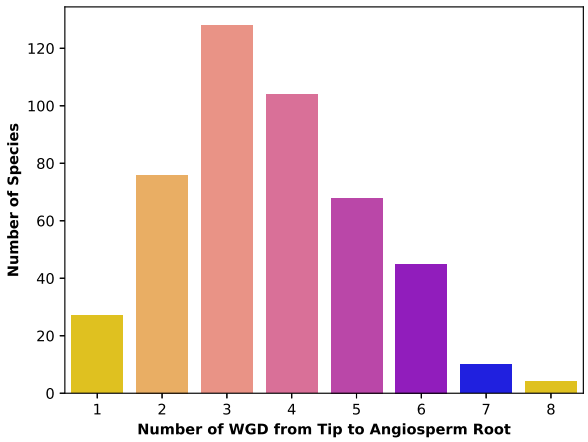


FIGURE 5 Histogram of the rounds of WGD in the history of each species included in the Janssens et al. topology. Rounds of WGD are based on the WGD hypotheses supported by WHALE models plus the AMBOα duplication shared by all angiosperms (Jiao et al., 2011; OTPT, 2019).

placement. Of the remaining 28 WGD hypotheses supported by MAPS that we could test, 23 were also supported by WHALE models (~82%) (Appendix S1: Table S6). Regardless of which analysis is considered, these inferences result in an average of 3.5 WGDs in the history of each angiosperm species when including the AMBOα duplication shared by all angiosperms (Figure 5) (Jiao et al., 2011; OTPT, 2019).

DISCUSSION

Our inferences of WGDs using state-of-the-art methods and two different species-tree topologies were largely consistent with previous analyses of WGDs across the angiosperms. We observed approximately 170 WGDs across our data set, compared to 179 reported in the OTPT (2019) analysis. These observations result in an average of ~3.5 WGD in the history of angiosperm species, regardless of method or tree topology used. Although KsRates, WHALE, and the OTPT (2019) analysis largely agreed on the relative number of WGDs inferred among lineages, their precise placement within the angiosperm phylogeny was variable. Approximately 80% of WGD hypotheses were inferred to be shared by the same species under both the TimeTree (Kumar et al., 2017, 2022) and Janssens et al. (2020) phylogenies. The remaining 20% of the WGD hypotheses unique to each species tree were largely a result of alternative species-tree topologies altering hypothesis generation. Given this degree of variation, and regardless of which WGD inference method employed, future researchers should carefully consider the limitations of methods used to infer species trees. Generally, WGD inference is dependent on the quality of the species-tree inference and, as we found, relatively small differences in tree topologies can significantly impact WGD hypotheses.

We observed that KsRates proposed an inflated number of WGDs (~245), many of which represented alternative hypotheses for particular WGDs. Variation in how we corrected synonymous substitution rate heterogeneity was the primary cause of alternative WGD hypotheses. Additional variation in WGD placement was also caused by multiple or alternative mixture models fit to the paralog distribution. Mixture models are known to overestimate the number of components (Tiley et al., 2018; Sensalari et al., 2022) and can be biased by peak shape variation due to gene conversion (Qiao et al., 2019; Sensalari et al., 2022) and allopolyploidy (Marcet-Houben and Gabaldón, 2015; Sensalari et al., 2022). For example, after accounting for rate heterogeneity within Brassicales, we observed a difference of ~0.3 in the median K_s of the ARTHα WGD compared to previous reports using uncorrected K_s values (Barker et al., 2009; Wang et al., 2013; Kagale et al., 2014; Jeong et al., 2016). Even minor variation in the estimate of mean K_s of ortholog and paralog peaks can result in shifting WGD hypothesis up or down branches. Variation in K_s inferences were particularly exaggerated among species-specific WGD, often shifting WGD up a node in the alternative tree topology. Continuing our example in Brassicales, our K_s inferences using the Janssens et al. (2020) phylogeny are largely consistent with prior reports of WGDs within this group (Barker et al., 2009; Wang et al., 2013; Kagale et al., 2014; Jeong et al., 2016; Mabry et al., 2020). However, the TimeTree (Kumar et al., 2017, 2022) topology was different from previously inferred phylogenies (Barker et al., 2009; Edger, 2012; Edger et al., 2018; Mabry et al., 2020; Walden et al., 2020) and resulted in two alternative placements for the ARTHβ duplication (Barker et al., 2009; Wang et al., 2013; Kagale et al., 2014; Jeong et al., 2016; Mabry et al., 2020) that were also supported by syntenic and WHALE analyses. These results indicate that although K_s methods are an accessible and powerful tool for assessing the presence of WGD within species, they should be used carefully when placing WGDs on a phylogeny.

Bayesian gene-tree to species-tree reconciliation approaches for estimating the rates of gene duplication and loss are relatively robust to topological variation when assessing if hypothesized WGD placements are accurate. We used WHALE as a post hoc test for WGD inferences made by KsRates, allowing us to reject a large fraction of spurious WGD hypotheses. This effort results in a similar number of WGDS (~170) supported by WHALE compared to previous reports using other methods (OTPT, 2019). However, the critical and relaxed WHALE models often disagreed on the precise placement of WGDs under different species-tree hypotheses. Our results suggest that duplication (D) and loss (L) rates among angiosperms are highly variable, and models that account for rate heterogeneity both among and within branches, such as the relaxed model, are less sensitive to topological variation than those that do not, consistent with prior reports (Zwaenepoel and Van de Peer, 2019a). In contrast, when holding duplication and loss rates constant within branches, WHALE models underestimated the number of WGD by almost 70%, and often rejected well

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known WGDs such as the ARTHα WGD in the Brassicales, which were supported in the work of Zwaenepoel and Van de Peer (2019a). Even when using the more robust relaxed model, 20% of hypothesized WGD placements were species tree specific, 76% of which were unique to single species. This disagreement was primarily caused by the initial hypothesis-generation step, as species-specific WGD inferences made by KsRates often varied under each species-tree topology. WHALE models also largely agreed with prior inferences from MAPS in the OTPT (2019) analyses. However, much like disagreements between WHALE models, a large fraction (~67%) of disagreements with previous MAPS analyses were due to species-tree topology differences between our study and the OTPT (2019) analyses. These results suggest that although the relaxed WHALE model appears to be reasonably robust to species-tree topological variation, it is subject to the limitations of initial hypothesis generation.

We used syntenic analyses as an alternative method to test hypothesized WGD placements. The availability of genomic data allowed us to test a total of 149 WGD hypotheses across our analyses, of which syntenic depth inferences supported 116. Notably, neither WHALE model supported a large fraction of these WGD hypotheses (~35%). Three of these hypotheses were also supported by MAPS; however, the remaining disagreements between the syntenic depth and WHALE analyses may be due to limitations of our syntenic depth heuristic. Our syntenic analyses often supported both alternative KsRates hypotheses for a given WGD, as observed for the ARTHβ WGD in the A1 analysis under the TimeTree topology (Kumar et al., 2017, 2022). Similarly, our syntenic depth heuristic was also sensitive to the species-tree topology, due to the inherent reliance on outgroup selection for making syntenic depth plots. Many tools already exist for analyzing complex syntenic data; however, they also require a priori knowledge of the shared WGD history among species (Tang et al., 2011; Parey et al., 2020; Siddiqui and Conant, 2023). Given the added power that syntenic analysis can give to disentangling complex WGD histories, our analyses and other WGD-focused syntenic tools highlight the need for formal tests to infer the shared polyploid ancestry of species using synteny. Similarly, synteny has recently been used to improve the inference of gene-tree and species-tree hypotheses by using syntenic clustering approaches and accounting for complex duplication and loss histories (Parey et al., 2020; Zhao et al. 2021). Species trees that are informed by the complex duplication history of plants may offer more robust hypotheses for placing WGD events. It is important to note that the genomes in our sample represent only one quarter of the total species present in our species trees. With the increasing availability of high-quality reference genomes, future efforts into disentangling the complex paleopolyploid ancestry of flowering plants will likely benefit from increased usage of synteny-based methods.

The topology of species-tree hypotheses is arguably more impactful on WGD inference than the use of different WGD inference methods. Some methods used in this study were more robust to variation than others; however, the most frequent cause of disagreements among methods in this study, as well as the results of prior work (OTPT, 2019), are topological differences between species trees. Here we have only evaluated WGD inferences using two alternative topologies, making it difficult to discern exactly how branching patterns and branch lengths respectively impact WGD inferences. Branch lengths directly determine both synonymous rate corrections and estimated rates of gene duplication and loss (Li et al., 2018; Zwaenepoel and Van de Peer, 2019a; Sensalari et al., 2022). However, it is unclear how sensitive WGD inferences that were made using these methods are to variation in branch lengths between alternative phylogenies. Error in the inferred evolutionary relationships among species likely represents an even larger challenge to WGD inference. It is well known that paralogy reconciliation and species-tree inferences influence each other (Shi, 2013; Struck, 2013; Smith and Hahn, 2022; Xiong et al., 2022), leading some to estimate them concurrently (Boussau et al., 2013; Szöllősi et al., 2014). Similarly, incomplete lineage sorting (Solís-Lemus et al., 2016; Yan et al., 2022), homeologous exchange (Liston et al., 2020), and violations of strict species-tree bifurcation (e.g., allopolyploidy; Oxelman et al., 2017; Liu et al. 2022) can complicate species-tree and gene-tree inferences. When using prior curated species trees to reconcile gene trees, estimations of duplication and loss rates are constrained to the species-tree topology and inherit any associated errors. Variation in species-tree topology likely impacts, potentially to different degrees, several other methods for inferring WGDs that rely on comparison to a species tree (Chen et al., 2000; Pfeil et al., 2005; Rabier et al., 2014; Li et al., 2015, 2018; Yang et al., 2015; McKain et al., 2016; Thomas et al., 2017; Zwaenepoel and Van de Peer, 2019a, 2020). Inferring accurate species trees has long been a central challenge in evolutionary biology (see Anderson et al., 2012; Szöllősi et al., 2014; Morel et al., 2022), and WGD inferences ultimately depend on a solid phylogenetic foundation. In our analyses, methods that average across studies, such as TimeTree, produced more spurious WGD inferences relative to trees made using traditional approaches that consider several single-copy gene families, such as the Janssens et al. (2020) phylogeny. Methods that consider multiple gene families under a coalescent model, such as ASTRAL (Boussau et al., 2013; Shekhar et al., 2017), that consider incomplete lineage sorting (ILS) (Shekhar et al., 2017, Yan et al. 2022), or account for complex duplication histories among gene families, such as SCORPiOs (Boussau et al. 2013; Parey et al. 2023), may provide a more robust scaffold for mapping WGDs. When data are available, the most robust methods for inferring the species tree should be used to provide a strong foundation for mapping WGD hypotheses.

Here we explored the distribution of ancient WGDs across angiosperms. We found that a variety of current methods are robust in detecting the presence of WGDs across lineages. However, we observed significant sensitivity to species-tree topology when placing WGDs within two

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angiosperm phylogenies. Given that species trees are hypotheses themselves, to which we map WGD hypotheses, it is critical to carefully assess how different WGD inference methods depend on the species tree. Many fundamental questions in polyploid evolution are beginning to be addressed with the increased availability of genomic data (Jiao et al., 2011; Vanneste et al., 2014; Tank et al., 2015; Soltis and Soltis, 2016; Landis et al., 2018; Smith et al., 2018; Bohutinska et al., 2020; Hao et al., 2021; Li et al., 2021; Qi et al., 2021; Marchant et al., 2022; Qiao et al., 2022; Almeida-Silva and Van de Peer, 2023; Bhadra et al., 2023; Blischak et al., 2023). Accurate inferences for shared polyploid ancestry among angiosperm species are necessary when forming predictions of polyploid plant evolution. For these reasons, future work should characterize the impacts of variation in species-tree topology on methods for inferring ancient WGDs. Methods such as syntenic analyses may offer additional evidence for disentangling complex WGD histories, both in aiding the construction of species trees in the presence of complex duplication histories and in making the WGD inferences. However, more formal methods are required, as our study highlights. As we expand the availability of data across the tree of life, the goal of accurate WGD inferences will likely reveal new insights into the profound impacts polyploidy has had on the evolution of plants and other lineages.

AUTHOR CONTRIBUTIONS

M.T.W.M. and M.S.B. designed the project, analyzed the data, and wrote the manuscript. G.F. assisted with preparing the Janssens et al. (2020) sub-analyses.

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DATA AVAILABILITY STATEMENT

Data and scripts used to analyze these data are available at the following git repository: https://gitlab.com/barker-lab/ajb_special_issue_2024.git.

OPEN RESEARCH BADGES

This article has earned Open Data and Open Materials badges. Data and materials are available at https://gitlab.com/barker-lab/ajb_special_issue_2023.git.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

Appendix S1. Supplemental tables.

Table S1. A list of reference genomes included in this study.

Tables S2, S3. KsRates results under the Janssens et al. (2020) and TimeTree topologies.

Tables S4, S5. Summary of WGD inference support for each method on each node in each sub-tree analysis for the Janssens et al. (2020) and TimeTree topologies.

Table S6. Summary of WGD inference support from each method at all the nodes in merged Janssens et al. (2020) topology.

Appendix S2. Supplemental figures.

Figure S1. Heat map of the bidirectional syntenic depth inferences made by MCScan for each species.

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Figures S2–S85. Phylogenies for each analysis under the TimeTree and Janssens et al. (2020) phylogenies. Each tree is annotated with hypothesized WGD placements supported by synteny, WHALE models, or MAPS. MAPS inferences were based on the equivalent MRCA for species inferred to share the proposed WGD in the OTPT (2019) analysis. Annotations are as follows: MAPS (red), WHALE critical model (purple), relaxed model (peach), and the syntenic depth test (blue).

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